Machine Learning Intelligent Chip Design

Final Project

NoC-based CNN Accelerator System with AXI4 DMA Architecture & Design Optimization

Description

In the Final Project, you are required to replace the ROM-based data storage used in HW4 with a DRAM-based memory model that supports both read and write operations. The accelerator must access data in DRAM through an AXI4-based DMA mechanism. On-chip SRAM or local SRAM may be used as optional buffering components.

In addition, you must propose an optimized design and compare it with the baseline design. The optimization is not limited to reducing execution cycles. You should clearly explain your design choices, evaluate their trade-offs, and discuss the advantages and disadvantages of your optimized design.

Implementation Details

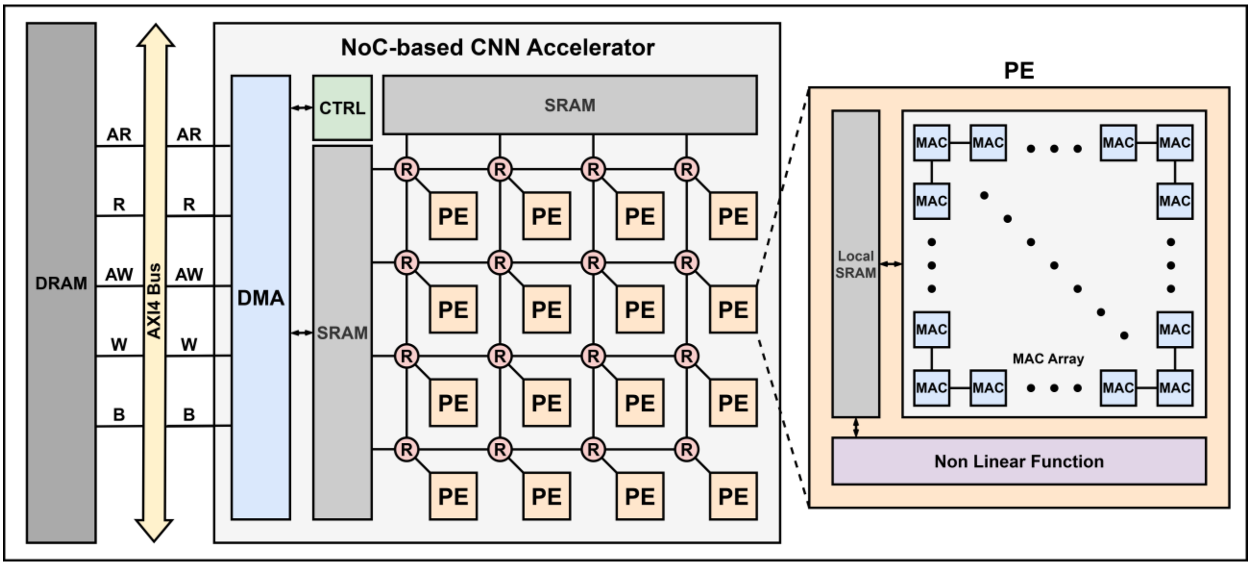

Figure 1. Overview of the NoC-based CNN accelerator system with AXI4 DMA.

In this project, the input images, weights, bias values, intermediate data, and output results should be stored in an external DRAM model instead of the ROM used in HW4. The DRAM must support both read and write operations. All data transfers between DRAM and the accelerator must be performed through an AXI4-based DMA mechanism. During computation, intermediate data that cannot fit in the on-chip SRAM or local SRAM may be written back to DRAM and later read again through the DMA.

You should define the accessible memory regions in DRAM, including the address ranges for inputs, weights, bias values, intermediate data, and output results. The detailed memory map may be designed by yourself, but it must be clearly described in the report.

The minimum required system must include a DRAM model, AXI4-based DMA access, a PE array with **at least 4x4 PEs**, a mesh-based NoC, and a controller that coordinates data movement and computation. The system must correctly execute the required test cases and print the correct classification results.

Inside the accelerator, you may use on-chip SRAM for data buffering and data reuse. You may also add local SRAM inside each PE if needed. However, using SRAM is not mandatory. If SRAM is used, it does not need to be implemented at the gate level, but its behavior model should be clearly defined, including its read/write behavior, latency, storage capacity, and data buffering behavior. The assignment does not impose a fixed upper limit on SRAM capacity or MAC resources, but these resources are part of the design trade-off and may affect the evaluation of the optimized design.

Except for replacing the ROM-based data access mechanism and modifying the controller to support AXI4 DMA-based memory access, the baseline system should follow the HW4 design. You may directly use your submitted HW4 files as the starting point for the Final Project.

For the optimized version, you may modify any part of the system. However, all off-chip memory accesses must still be performed through the AXI4 DMA-based memory access mechanism.

Please note that the parameter files inside the data folder are the only files that must not be modified. **The data folder must remain unchanged.**

Minimum Requirements and Design Rules

1. Both the baseline and optimized designs must keep at least a 4x4 mesh-NoC. The optimized design may modify the PE internal architecture, MAC resources, local SRAM, dataflow, routing policy, buffering strategy, and scheduling method, but the physical NoC size must not be smaller than 4x4.
2. The DRAM layout may be designed by each student. The memory map must include inputs, weights, bias values, intermediate data, and output results, and must be clearly described in the report.
3. The output results must be written back to DRAM before being printed or checked.
4. The data folder may only be used to initialize the DRAM model. During CNN execution, the controller, DMA, PE, NoC, and accelerator modules must not directly read image files, weight files, or bias files from the data folder. All accelerator data accesses during computation must go through the DRAM model and the AXI4-based DMA mechanism.
5. Both the baseline and optimized designs must print the correct top-100 classification results for make cat and make dog. The top-100 classification results must be exactly correct.
6. Both FP/ and FP_Optimized/ must contain a complete and independently compilable SystemC project. The TA will enter each folder and run make cat and make dog.

Implement Notes

You may refer to the provided **AXI_introduction.pptx** for details about the AXI4 protocol. The AXI4-based DMA model must at least support burst transfers, VALID/READY handshake behavior, transfer length information such as ARLEN/AWLEN, and transfer size information such as ARSIZE/AWSIZE. A simplified AXI4 model is acceptable, but the implemented signals, parameters, timing behavior, and abstraction level must be clearly described in the report. Supporting outstanding transactions is optional. Designs that implement a more complete AXI4 behavior, such as five independent AXI channels and outstanding transactions, may receive higher scores in the AXI completeness evaluation.

For the optimization part, you may optimize any part of the system. The optimization does not have to focus only on reducing the total number of execution cycles, and the accelerator (simulator) does not have to be fully cycle accurate. You may also consider other design objectives, such as reducing the number of MAC units, reducing SRAM capacity, reducing DRAM or SRAM access volume, improving NoC utilization, improving PE utilization, reducing NoC transfer latency, improving data reuse, redesigning the dataflow, supporting multicast, overlapping computation and communication, or improving the cycle accuracy of the simulator. SRAM capacity and MAC resources are not strictly limited, but using more resources should be justified through a clear trade-off analysis.

Your report must clearly explain the optimization goal, design method, trade-offs, and evaluation metrics. For example, an optimized design may use fewer MAC units at the cost of longer execution time, reduce DRAM accesses by increasing SRAM capacity, improve NoC efficiency through multicast, or improve data reuse through a different dataflow. Your implementation must be consistent with the design described in your report. **Any form of falsification will result in a zero score.**

You do not need to download or extract any additional files. **You may directly modify your HW4 files to complete the Final Project.**

Output Result Format

After running the make cat and make dog commands, your program must print the classification results in the same format as the previous assignments. You may also print additional information for analyzing your optimized design, such as execution cycles, memory access counts, PE utilization, NoC statistics, SRAM usage, or SystemC execution time.

Submission Guidelines & Grading Policy

- The grading breakdown for this assignment is as follows:

Functionality: 30%

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Both FP/ and FP_Optimized/ must integrate a DRAM model, AXI4-based DMA access, at least a 4x4 PE mesh-NoC, and a controller for data movement and computation. After executing make cat and make dog, both versions must display the exactly correct top-100 classification results.
You may also print additional information to help analyze your optimized design, such as execution cycles, memory access counts, PE utilization, NoC transfer statistics, SRAM usage, or SystemC execution time.

Report & Optimized Design: 70%

You must propose and implement an optimized design and compare it with the baseline design. The optimization may involve trade-offs among computation resources, memory hierarchy, data movement, NoC communication, simulation accuracy, and dataflow design. Excessive resource usage may affect the trade-off evaluation and the optimized-design score.

The report and optimized method will be evaluated using the following rubric:

a. AXI completeness: 5%

Whether the AXI4-based DMA design properly supports AXI-like behavior, including five independent AXI channels, burst transfer, VALID/READY handshake, transfer length, transfer size, and optional outstanding transactions.

b. SRAM model and usage: 10%

Whether the design includes a reasonable SRAM behavior model and uses SRAM effectively for buffering, data reuse, or **reducing DRAM access**. SRAM is optional for functionality, but it is evaluated as part of the optimized design score. Designs without SRAM may still be functionally correct, but cannot receive full credit in this item.

c. Trade-off analysis and evaluation table: 10%

Whether the report clearly analyzes the trade-offs among execution cycles, memory access volume, SRAM capacity, MAC resources, NoC utilization, PE utilization, and other metrics.

d. Optimization implementation: 15%

Whether the optimized design is actually implemented in code and is consistent with the report, rather than only described conceptually.

e. Memory layout design: 5%

Whether the DRAM address map and data placement are clearly defined, reasonable, and consistent with the implemented dataflow.

f. Evaluation metrics and baseline-vs-optimized comparison: 5%

Whether the baseline and optimized designs are compared using meaningful metrics and tables.

g. Abstraction level and computation modeling: 10%

Whether the computation model reasonably reflects hardware behavior. Designs that model MAC resource sharing, scheduling, and data movement more explicitly will receive higher scores than designs that simply unroll all computations using software-like for loops without meaningful hardware abstraction.

h. Report quality: 10%

Whether the report is clear, complete, well organized, and consistent with the submitted source code.

If the source code is not submitted, the report will receive no credit.

Required Evaluation Table

You must collect or estimate the selected metrics for both the baseline and optimized designs using a consistent measurement method. The comparison between the baseline design and the optimized design must be presented in tables. The following table is a suggested template. The metrics listed in the table are examples of possible evaluation items; you do not need to implement every technique listed here or fill in every row if it is not applicable to your design. If your optimization targets other aspects, you may add new rows or create additional tables.

Metric / Design Aspect	Baseline Design	Optimized Design	Notes / Trade-off
Number of PEs			
Number of MAC units in each PE			
Total number of MAC units			
PE utilization			
Local SRAM in each PE and its capacity			
Total on-chip SRAM capacity			
SRAM bit width			
Number of SRAM banks			
DRAM read volume and DRAM write volume (GB)			
DRAM bit width			

Metric / Design Aspect	Baseline Design	Optimized Design	Notes / Trade-off
Number of DRAM banks			
SRAM read volume and SRAM write volume (GB)			
Total memory access volume (GB)			
Execution cycles			
SystemC execution time			
NoC routing method			
Multicast support			
NoC protocol (AXI / ABP)			
NoC utilization			
NoC transfer latency			
Dataflow design			
Time breakdown of different operations			
Overlap of computation, communication, and memory access			
Other design-specific metrics			

Your report should explain why the selected metrics are important and how they reflect the trade-offs of your optimization. A design with fewer execution cycles is not always better if it requires significantly more MAC units, larger SRAM capacity, wider memory interfaces, more memory banks, higher memory access volume, lower PE utilization, or a more complex communication mechanism. Similarly, using more SRAM or more MAC units is allowed, but the benefit must be clearly justified in the evaluation table and discussion.

- For the file submission, you need to upload a zip archive to New E3. The file should be named FP_mlchipXXX.zip, where XXX is your account ID. The zip file must contain two versions of your code and the report. The report file should be named report_mlchipXXX.pdf, where XXX is your account ID. The internal structure of the zip file should follow the format shown in the figure below.

```
FP_mlchipXXX/
├── FP
│   └── Your design
├── FP_Optimized
│   └── Your optimized design
└── report_mlchipXXX.pdf
```

- If the submitted archive violates the naming rules or the required file structure, 10 points will be deducted.
- **The data folder and all of its contents, including input images, weights, and bias files, must not be included in your submission.**
- Make sure your code is well commented and clearly organized.
- **The following rules must be strictly followed. Any violation will result in 0 points.**
  - **Plagiarism is forbidden.**
  - **Chinese characters are not allowed anywhere in the submitted files, including comments.**